SAT MATH

Algebra • Skill 1.1 • Notes (filled in)

Skill 1.1 • Linear Equations in One Variable

Notes (filled in)

Learning goals

- Solve linear equations, including equations with parentheses, fractions and decimals.
- Decide whether a linear equation has one solution, no solution, or infinitely many solutions.
- Find the value of an expression without solving for x first, and write equations from word problems.

Vocabulary. linear equation, solution, distributive property, least common denominator, no solution, infinitely many solutions.

1. The strategy

Key idea

To solve a linear equation: (1) __distribute__ to remove parentheses; (2) multiply by the __least common denominator__ to clear fractions; (3) collect the x-terms on one side and the __constants__ on the other; (4) __divide__ by the coefficient of x. Check by substituting.

Example 1 • Parentheses

Solve $4(x-3)+5=2x+9$.

$4x-12+5=2x+9 \Rightarrow 4x-7=2x+9 \Rightarrow 2x=16 \Rightarrow x=8$. Check: $4(5)+5=25$ and $2(8)+9=25$.

Example 2 • Fractions

Solve $\dfrac{x}{3}+\dfrac{x+1}{4}=9$.

Multiply by 12: $4x+3(x+1)=108 \Rightarrow 7x+3=108 \Rightarrow 7x=105 \Rightarrow x=15$.

2. How many solutions?

Key idea

After simplifying to the form $ax+b=cx+d$:

- if $a \neq c$: exactly __one__ solution;
- if $a=c$ and $b \neq d$: __no__ solution (the x-terms cancel and a false statement remains, like $5=7$);
- if $a=c$ and $b=d$: __infinitely many__ solutions (both sides are identical).

Example 3

For which value of k does $3(2x+k) = 6x+12$ have infinitely many solutions? For which values does it have no solution?

$6x + 3k = 6x + 12$. The x-terms are equal, so we need $3k = 12$: $k = 4$ gives infinitely many solutions. Any $k \neq 4$ gives no solution.

3. Find the expression, not x

The SAT often asks for an expression such as $10x - 4$. Look for a multiple of the given equation.

Example 4

If $5x - 7 = 13$, what is the value of $10x - 4$?

Multiply by 2: $10x - 14 = 26$, so $10x - 4 = 26 + 10 = 36$. (Or: $x = 4$, then $10(4) - 4 = 36$.)

4. Word problems

Example 5

A plumber charges a \$65 call-out fee plus \$40 per hour. A bill was \$245. How many hours did the plumber work?

$65 + 40h = 245 \Rightarrow 40h = 180 \Rightarrow h = 4.5$ hours.

AP exam tip

[SAT tip] Use Desmos when an equation is messy: graph $y =$ left side and $y =$ right side and read the x-coordinate of the intersection. Parallel lines mean no solution; the same line means infinitely many.

Common mistakes

- Distributing to only the first term: $-2(x-3) \neq -2x - 3$; it is $-2x + 6$.
- Multiplying only some terms by the common denominator.

Check your understanding

1. $7x - 4 = 3x + 20$ gives $x =$ 6.
2. $2(x+1) = 2x + 3$ has no solution.
3. If $3x + 7 = 22$, then $6x - 1 =$ 29.